Fertility and Housing Tenure Choice

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Macroeconomics Student Lunch

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Housing and Fertility

≡ Newsweek Nation World Lifestyle Opinion Rankings

News Birth Rate Population Demographics Population Growth Housing Housing Ma

News Article

US Birth Rate Warning Issued Over House Prices

Housing Market | Real Estate 4 minute read

US birth rate hits record low, housing costs weigh on family planning

The 2008 financial crisis pushed both housing and fertility

POLICY

Will cheap housing lead to more babies?

The connection between housing and fertility rates has a missing piece.

Want to Boost Birthrates? Build More Homes

Policymakers looking to reduce childcare costs should work to expand housing supply.

Renting a home ‘leads to lower birth rates’

Housing, Fertility, and Tenure Choice

Housing costs affect fertility—but tenure and wealth matter

- Owners have more kids when house prices or housing wealth rise; renters do not (Dettling & Kearney 2014), and big units matter (Couillard 2025)

Children increase how big of a house you need. This has implications for:

- Owning: tenure segmentation
- Location: housing cost differs geographically
- Wealth: borrowing constraints, bequests

Physical space is key. How much does it affect fertility?

What kind of housing policy is good for natality?

$\rightarrow$ focus on space constraints and tenure choice

This Paper

Need a structural model where tenure, location, and fertility are joint. Today:

Model: lifecycle model of fertility, location, tenure, and saving

- Childbirth is a discrete shock to housing need
- Housing markets with borrowing constraints, tenure choice, transaction costs
- Location and tenure respond to family-space need

Empirics: document facts about fertility, housing, and moving

- ACS: parents less central; re-sort toward periphery within metros
- PSID: first birth raises ownership, rooms, and moves for space

Goal: write a model disciplining these interactions, think about policy.

Model

Environment

- Two locations: **center** (C) and **periphery** (P) within a metro area
  - Center has an amenity premium but less elastic housing supply
  - Common wages; sorting driven by amenities, housing costs, and moving frictions
- Adult age $a \in \{0, 1, \ldots, J\}$; deterministic aging and death
  - Working age $a \in [0, J_R)$; fertile window $a \in [J_f^{\text{start}}, J_f^{\text{end}}]$
  - Children do not work; they depend on their parents
- Households choose:
  - **Fertility**: $n' \in \{0, 1, \ldots, \bar{n}\}$, subject to EV shocks
  - **Location**: center or periphery each period (with moving costs and EV shocks)
  - **Tenure**: rent or buy a discrete house size H_k on a housing ladder
  - **Saving**: liquid wealth b with collateral borrowing for owners
- Children age stochastically through life stages; housing need depends on children
- Benchmark city scale is normalized to 1; counterfactual city size responds to entry from city-born and outside-origin young adults

Preferences

n: fertility; s: child-aging stage (stochastic); $m = m(n, s)$: children currently at home.

Flow utility. Over consumption and housing services:

$$u(c, \mathbf{h}; m) = \frac{\left[\left(c - \bar{c}(m)\right)^{\alpha}\left(\mathbf{h} - \bar{h}(m)\right)^{1-\alpha}\right]^{1-\sigma}}{1-\sigma} + \psi_{\text{child}}\, m.$$

Subsistence requirements. Depend on family size:

$$\bar{c}(m) = \bar{c}_0 + \bar{c}_1 m, \qquad \bar{h}(m) = \bar{h}_0 + \underbrace{\bar{h}_{\text{jump}} \mathbf{1}\{m \geq 1\}}_{\text{first-child housing margin}} + \bar{h}_1 m.$$

Housing services. Households either rent or own; owners get a service premium:

$$\mathbf{h} = \underbrace{\mathbf{1}\{\text{own}\}\, \chi\, h}_{\text{owner services}} + \underbrace{\mathbf{1}\{\text{rent}\}\, h^R}_{\text{renter services}}, \quad \chi > 1.$$

- Child costs disappear after children mature.

Bequests

Terminal utility at death:

$$\mathcal{B}(\widetilde{b}, n) = \theta_0 \big(1 + \theta_n n\big) \frac{(\theta_1 + \widetilde{b})^{1-\sigma}}{1-\sigma}, \qquad \widetilde{b} = b + (1-\psi) p_i h.$$

- θ_0 governs overall bequest motive (saving level).
- θ_n governs the parent–childless saving differential.
- Total bequeathable wealth $\widetilde{b}$ includes housing equity net of the transaction cost for owners.

Housing, Tenure, and Budget Constraints

Renters choose housing continuously:

$$h^R \in [0, \bar{h}^R].$$

Owners choose a discrete house size:

$$h \in \{H_1, \dots, H_{N_H}\}.$$

Renters

$$b' = Rb + y(a) - c - r_i h^R, \qquad b' \geq 0.$$

Owners

$$b' = Rb + y(a) - c - (\delta + \tau_H) p_i h' + \underbrace{\mathbf{1}\{\text{sell}\}(1-\psi) p_i h}_{\text{sale proceeds}} - \underbrace{\mathbf{1}\{\text{buy}\}\, p_i h'}_{\text{purchase cost}}, \qquad b' \geq -\phi p_i h'.$$

- **Tenure segmentation**: renters access housing up to $\bar{h}^R$; owner sizes start at $H_1 > \bar{h}^R$.
- Movers and newly matured children arrive as renters, then re-choose tenure and size.

Fertility and Child Aging

- Fertility choices are made only in the fertile window $a \in [J_f^{\text{start}}, J_f^{\text{end}}]$.
- One-shot fertility: can have multiple kids, but choose only once:

$$n' \in \{0, 1, \ldots, \bar{n}\}.$$

- After birth, children age through stages s (think toddler $\rightarrow$ child $\rightarrow$ teen $\rightarrow$ adult):
  - Aging is **sequential**: each stage transitions only to the next one, with random arrival
  - All n' children of a parent age **together** as a single cohort - s is shared.
- While children are at home ($s <$ matured), they raise housing need $\bar{h}(m)$ and consumption cost $\bar{c}(m)$.
- Once matured, child costs shut off and children enter the economy at age $a = 0$.

Firms and Earnings

- Competitive firms produce with labor:

$$Y = AL, \qquad w = A.$$

- **Common wages** across locations: sorting driven entirely by amenities and housing costs.
- Age-dependent earnings:

$$y(a) = \begin{cases} (1 - \tau_{\mathrm{pay}})\, w(a) & a < J_R \\ \bar{b} & a \geq J_R \end{cases}$$

- Retirees receive a stationary pay-as-you-go pension $\bar{b}$ financed by payroll taxes.

Housing Supply and Price Mapping

Housing supply in each location:

$$H_i^s(r_i) = \bar{r}_i r_i^{\eta_i}, \qquad i \in \{C, P\}.$$

- Less elastic center supply makes family space more expensive in the core.
- House prices map into user costs through

$$r_i = (R + \delta + \tau_H) p_i.$$

Young-Adult Entry: Environment

- Incumbent households live in the center or periphery, choose housing, savings, location, and fertility, and their children age inside the household.
- When children mature, they become city-born potential young adults. At city scale S, their flow is
$$SB_0.$$
- Each period there is also an outside-origin pool of potential young adults, with mass M.
- Both pools face the same entry choice:
$$C \quad \text{(center)}, \qquad P \quad \text{(periphery)}, \qquad E \quad \text{(outside economy)}.$$

Young-Adult Entry: Choice

- Potential entrants are outside-origin young adults plus maturing city-born children:
$$M + SB_0.$$
- Each potential entrant draws entry tastes and chooses center, periphery, or the outside economy.
- Entering modeled location $i \in \{C, P\}$ means starting adult life as a childless renter, $x_i^E = (b^E, R, i, 1, 0, 0)$, with value W_i^E. The outside economy gives value $\bar{W}^E$.
- These values imply probabilities:
$$\pi_C^E + \pi_P^E + \pi_{\text{out}}^E = 1, \qquad q^E = \pi_C^E + \pi_P^E = 1 - \pi_{\text{out}}^E.$$
- The potential entrant mass is then allocated as
$$E_i = (M + SB_0)\pi_i^E, \quad i \in \{C, P\}, \qquad E_C + E_P = (M + SB_0)q^E.$$

Stationarity by Location

Accounting

- Let g be the normalized stationary composition. At city scale S, level masses are given by Sg.
- Let E_{0i} be the per-unit mass of age-one households that must enter location i to reproduce the stationary composition:

$$E_0 = E_{0C} + E_{0P}.$$

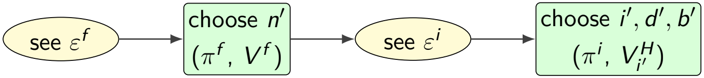
- At scale S, the potential entrant pool is $M + SB_0$. Full stationarity requires one condition for each modeled location:

$$SE_{0i} = [M + SB_0]\,\pi_i^E, \qquad i \in \{C, P\}.$$

- Summing the two location equations gives the aggregate scale condition:

$$SE_0 = q^E\,[M + SB_0]\,, \qquad S = \frac{q^E M}{E_0 - q^E B_0}.$$

The Household's Problem



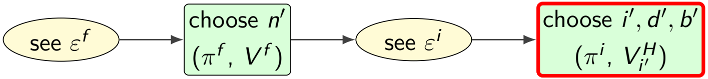
The full household state is

$$x = (b, d, i, a, n, s).$$

- b: liquid wealth; d: tenure; i: location; a: age.
- n: fertility; s: child-aging stage.
- After the fertility choice, n' is the period's fertility state; if no birth, $n' = n$.

Within the period: choose n' and then choose a destination–housing–saving bundle (i', d', b').

The Household's Problem: Housing and Saving



For each candidate destination i', the household chooses tenure and next assets:

$$V^H_{i'}(b, d, i, a, n', s) \;=\; \max_{d',\, b'} \left\{ u\big(c, \mathbf{h}(d'); m(n', s)\big) + \beta\, \mathbb{E}_{s'}\left[V(b', d', i', a+1, n', s') \right] \right\}$$

Let $H(R) = 0$, $H(O_k) = H_k$, and $\tau = \mathbf{1}\{(i', d') \neq (i, d)\}$. Feasible branches:

- **Rent:** $d' = R$, $\mathbf{h}(d') = h^R \in [0, \bar{h}^R]$,

$$c + b' + r_{i'} h^R = Rb + y(a) + \mathbf{1}\{d \neq R\}(1-\psi) p_i H(d),$$
$$b' \geq 0.$$

- **Own:** $d' = O_k$, $\mathbf{h}(d') = \chi H_k$,

$$c + b' + (\delta + \tau_H) p_{i'} H_k + \tau p_{i'} H_k = Rb + y(a) + \tau \mathbf{1}\{d \neq R\}(1-\psi) p_i H(d),$$
$$b' \geq -\phi p_{i'} H_k.$$

The Household's Problem: Location Choice

see ε^f → choose n' $(\pi^f,\ V^f)$ → see ε^i → choose i', d', b' $(\pi^i,\ V^H_{i'})$

After fertility choice n', draw destination taste shocks $\varepsilon^i_{i'}$ and choose a destination. Let

$$\widetilde{V}_{i'} = V^H_{i'}(b, d, i, a, n', s) + E^R_{i'} - \mu_{ii'}.$$

The location object is the expected maximum over those shocks:

$$\begin{aligned} V^i(b, d, i, a, n', s) &\equiv \mathbb{E}_{\varepsilon^i} \max_{i' \in \{C,P\}} \left\{ \widetilde{V}_{i'} + \varepsilon^i_{i'} \right\} \\ &\propto \kappa_i \ln \sum_{i' \in \{C,P\}} \exp\left(\frac{\widetilde{V}_{i'}}{\kappa_i} \right). \end{aligned}$$

- $V^H_{i'}$ maximizes over d' and b' in destination i'.

Choice probabilities:

$$\pi^i_{i'}(b, d, i, a, n', s) = \frac{\exp\left(\widetilde{V}_{i'}/\kappa_i\right)}{\sum_{j \in \{C,P\}} \exp\left(\widetilde{V}_j/\kappa_i\right)}.$$

The Household's Problem: Fertility Choice

see ε^f → choose n' $(\pi^f,\ V^f)$ → see ε^i → choose i', d', b' $(\pi^i,\ V^H_{i'})$

Childless households in the fertile window choose fertility once:

$$V(b,d,i,a,0,0) \equiv \mathbb{E}_{\varepsilon^f} \max_{n' \in \mathcal{N}} \left\{ V^i(b,d,i,a,n',s(n')) + \varepsilon^f_{n'} \right\}$$

$$\propto \kappa_n \ln \sum_{n' \in \mathcal{N}} \exp\left(\frac{V^i(b,d,i,a,n',s(n'))}{\kappa_n} \right).$$

Choice probabilities:

$$\pi^f_{n'}(b,d,i,a,0,0) = \frac{\exp\left(\frac{V^i(b,d,i,a,n',s(n'))}{\kappa_n}\right)}{\sum_{n''} \exp\left(\frac{V^i(b,d,i,a,n'',s(n''))}{\kappa_n}\right)}.$$

Sorting and Housing-Demand Elasticity

$\mathcal{E}_C > \mathcal{E}_P$ (amenity); $r_C > r_P$ (rents); $M = Rb + y - b'$ (expenditure budget);
$\bar{h}(n)$ housing requirement with n children, $\bar{h}(1) > \bar{h}(0)$. Ignoring moving costs and Gumbel

Sorting.

- Higher $\bar{h} \Rightarrow$ larger rent bill in $C \Rightarrow$ preference for C falls.
- Single crossing: for each M, unique threshold $\hat{h}(M)$ s.t. $\bar{h} < \hat{h} \Rightarrow C$, $\bar{h} > \hat{h} \Rightarrow P$.
- $\hat{h}$ rises with M (richer households absorb high r_C).
- Since $\bar{h}(1) > \bar{h}(0)$: parents sort weakly peripheral.
  - Parent share falls with local rent.
  - First birth triggers moves to lower-r areas.

Elasticity. Housing demand for uncapped renter splits into committed + discretionary:

$$h^*(n) = \underbrace{\alpha\,\bar{h}(n)}_{\text{committed}} + \underbrace{\frac{(1-\alpha)(M-\bar{c})}{r}}_{\text{discretionary}}.$$

- Parents less income-elastic on intensive margin; less responsive to rent.
- Kids raise the probability of owning.

Sorting and rents

- In this static derivation, let $\mathcal{V}_i(n; M)$ be indirect utility for a renter with n children in location i.
- $\Delta\mathcal{V}(n; M) \equiv \mathcal{V}_C(n; M) - \mathcal{V}_P(n; M)$
- Sorting wedge: $\mathcal{W}(M) \equiv \Delta\mathcal{V}(0; M) - \Delta\mathcal{V}(1; M) > 0$
- If the rent gap $r_C - r_P$ widens, the parent-childless gap in location preferences $\mathcal{W}$ widens:

$$\left.\frac{\partial \mathcal{W}}{\partial r_C}\right|_{r_P} > 0.$$

- Across cities: the parent-childless center-share gap rises with the local center-periphery rent gap.

Constrained Squeeze

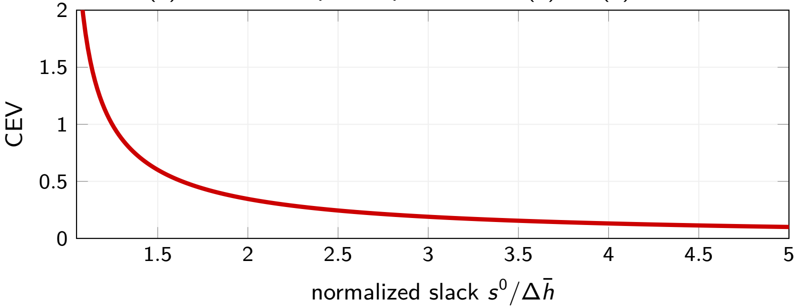
derivation

Capped renters and owners on a fixed rung cannot adjust housing. How costly is a child on a home that barely fits one?

Consumption-equivalent variation, holding housing fixed:

$$u(c + \mathrm{CEV},\ h,\ 1) \ = \ u(c,\ h,\ 0).$$

Pre-birth slack $s^0 = h - \bar{h}(0)$ and extra required space $\Delta\bar{h} = \bar{h}(1) - \bar{h}(0)$.



Summary

- A model of fertility, location, tenure, and saving with child-driven housing need shocks
- Children raise housing need, with implications for tenure and location choice
- The sorting wedge between parents and childless rises with the local rent gap.
- The closer a household is to being space-constrained, the more costly a child is
- Some predictions:
  - Parents sort peripheral, are less housing-elastic, and are more likely to own.
  - Sorting strengthens with the local rent gap.

  The data can help:
  - Improve our understanding of housing and fertility
  - Validate the predictions
  - Inform our quantification

Empirics

Data

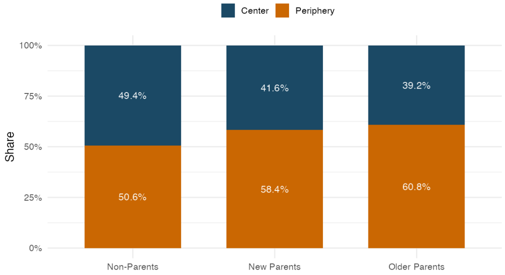
- **American Community Survey (ACS) 1-year estimates:**
  - Annual survey data collected by the U.S. Census Bureau
  - Geographic detail up to PUMA level
  - Demographic characteristics, socioeconomic indicators, and housing information

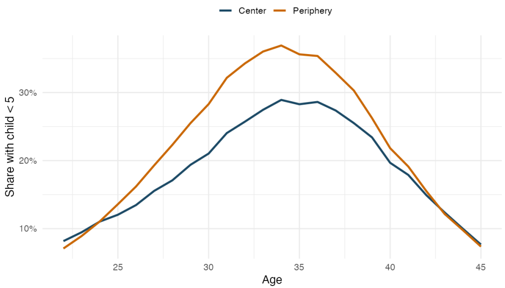
- **Panel Study of Income Dynamics (PSID):**
  - Longitudinal household survey initiated in 1968 (24,000 households in 2021)
  - Follows original sample families and their descendants across generations
  - Rich demographics, income and wealth information
  - Just got the geocoded data, so I can track location and moves at the zip level: next time!

Parenthood and Location

Geography

Model: parents sort to the periphery



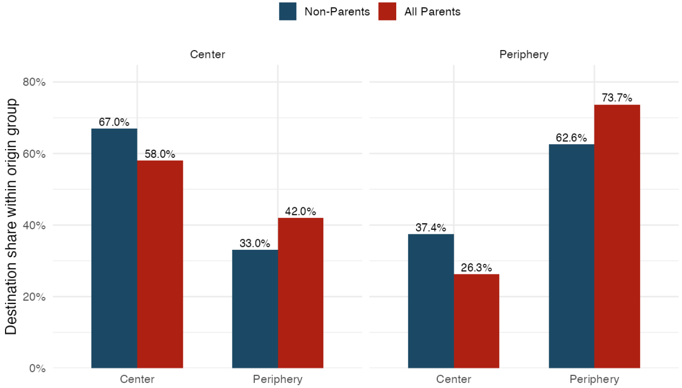


- Non-parents are more concentrated in the center than new parents: 0.494 vs. 0.416.
- More parents outside the center

From Movers: Parents Re-Sort Within Metro Areas

Regressions Within moves

Model: parents sort to the periphery; childbirth triggers moves to lower-r areas.



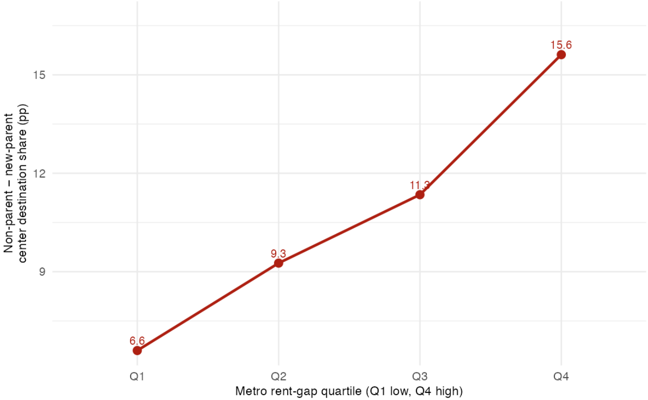
- Among within-metro movers, parents are less likely to choose center destinations and more likely to choose peripheral destinations.

  Baseline center shares: 49.4% for non-parents, 41.6% for new parents. Owner rates: 49% vs. 59%.

Sorting Strength and the Local Rent Gap

Regression Elasticity Stock

Model: parent share falls with local rent; $\partial\mathcal{W}/\partial r_C > 0$.



Y-axis: non-parent minus new-parent center destination share among within-metro movers (pp). Base destination shares: 52.4% for non-parents, 41.5% for new parents.

- The sorting gap among movers more than doubles from Q1 to Q4.

Using the PSID: Event-Study Design

For household i, let E_i denote the year of childbirth. For any housing outcome Y_{it}, estimate

$$Y_{it} = \sum_{k=-5, k \neq -2}^{10} \alpha_k \mathbf{1}\{t - E_i = k\} + \zeta_i + \tau_t + \delta X_{it} + \varepsilon_{it}.$$

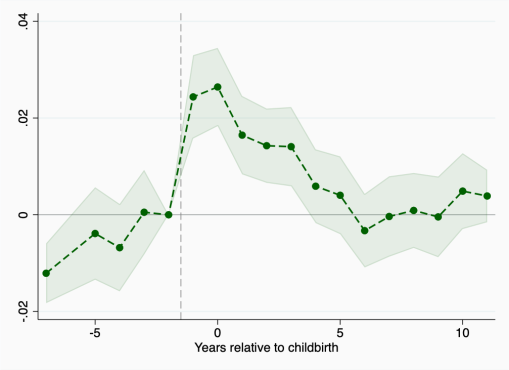
- ζ_i and τ_t are household and calendar-year fixed effects.
- The omitted period is $k = -2$, so each α_k is relative to two years before birth.
- I use the same design for homeownership, moving-for-size, and rooms.

PSID: Childbirth Raises Moves for Space

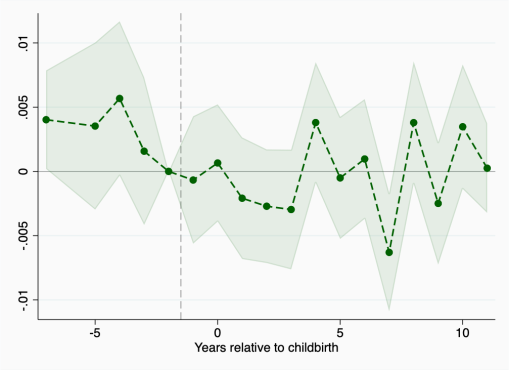
Reasons By Category No IDFE

$\hat{\alpha}_k$: change in probability of moving for reason Y, relative to $k = -2$.

$Y = \mathbf{1}\{\text{moved for expansion of housing}\}$



$Y = \mathbf{1}\{\text{moved for neighborhood quality}\}$



- The probability of moving for housing size rises around first birth.
- By contrast, moves for neighborhood quality do not rise in the same way.

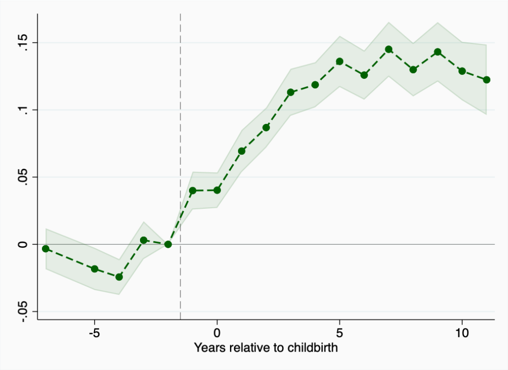
PSID: Ownership and Housing Size Around First Birth

Own: No IDFE Rooms: No IDFE

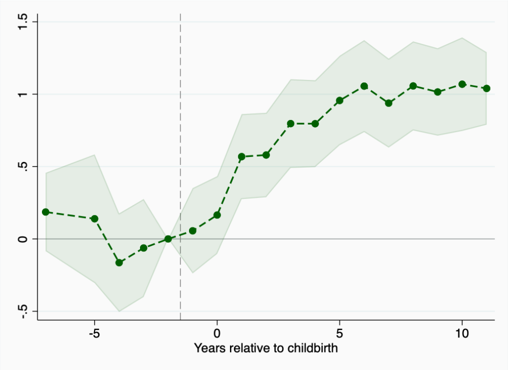
Model: $\bar{h}(1) > \bar{h}(0)$; *kids raise ownership.*

$\hat{\alpha}_k$: change relative to $k = -2$.

$Y_{it} = \mathbf{1}\{\text{owner}\}_{it}$



$Y_{it} = \text{number of rooms}_{it}$



- Homeownership rises sharply after first birth; rooms expand by about 0.66 at $k = +3$.

Pre-birth mean: 6.0 rooms.

Quantification

Externally Calibrated Parameters

	Param	Description	Value	Source
Pref.	σ	Risk aversion	2.0	–
	α	Consumption weight	0.70	BLS CEX
	$\bar{c}_0$	Consumption floor	0.10	Poverty line
Fin.	r	Interest rate	0.04	–
	δ	Depreciation	0.02	Greaney et al. (2024)
	τ_H	Property tax	0.01	US average
	ϕ	LTV limit	0.80	20% down payment
	ψ	Transaction cost	0.06	Greaney et al. (2024)
Life	J, J_R	Lifespan, retirement	60, 47	Ages 18–77, ret. 65
	τ_{pay}	Payroll tax	0.179	Greaney et al. (2024)
Housing	η_P, η_C	Supply elasticities	2.0, 0.8	Saiz (2010)

Internally Calibrated Parameters

Strategy

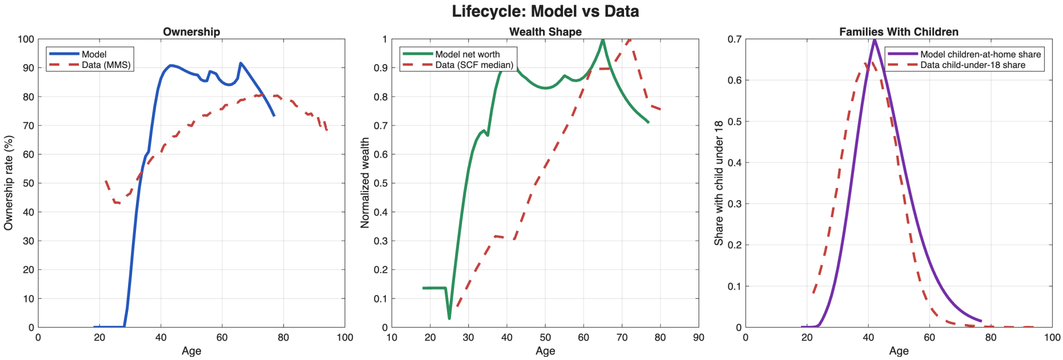
Param	Description	Value
$\mathcal{E}_C$	Center amenity	0.30
$\bar{r}_C$	Center rent anchor	0.09
β	Discount factor	0.935
b_0	Entry wealth	0.122
χ	Owner utility premium	1.047
ψ_{child}	Flow utility of children	0.067
κ_f	Fertility Gumbel scale	2.33
$\bar{h}_0$	Baseline housing need	2.52
$\bar{h}_{\text{jump}}$	Housing jump (1st child)	0.622
$\bar{h}_n$	Add'l child housing cost	0.909
$\bar{c}_n$	Add'l child cons. cost	0.092
κ_i	Location-choice Gumbel scale	1.50
μ_{move}	Moving cost	5.00
θ_0	Bequest shifter	0.530
θ_n	Bequest parent premium	0.250

Calibration Targets

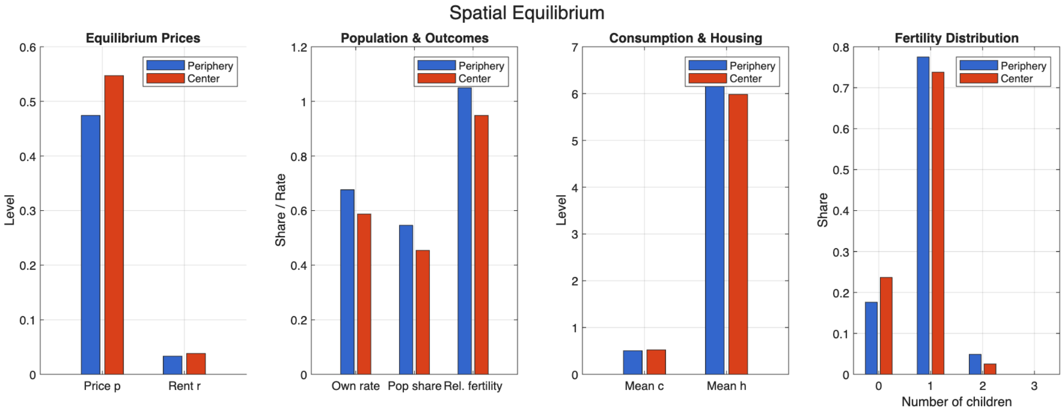
	Moment	Target	Model
Fertility	TFR	1.70	1.66
	Ever childless	0.15	0.21
	Age at first birth	26.0	34.8
	Fertility gradient $P-C$	0.133	0.168
Housing	Ownership, ages 30–55	0.627	0.751
	Ownership gradient $P-C$	0.170	0.185
	H_{01} (PSID, rooms)	0.664	0.323
	H_{12} (PSID, rooms)	0.566	0.299
	Parent ownership gap	0.110	0.184
Saving	Liq. wealth / income	0.60	0.460
Location	Center population share	0.450	0.454
	Rent/room ratio C/P	1.14	1.154
	Center share, non-parents	0.494	0.418
	Center share, new parents	0.416	0.392
	C–P switching rate	0.032	0.027
Bequest	Ownership, ages 65–75	0.863	0.850
	Parent–childless own gap	0.070	0.052

Lifecycle Fit

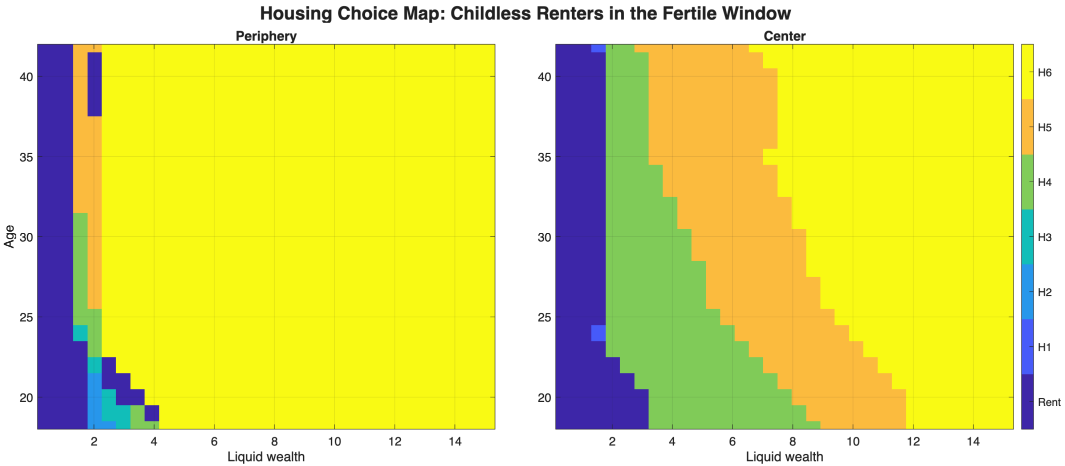
Housing Fit



Equilibrium



Housing Choice Map in the Fertile Window



Conclusion

- A model of fertility with realistic housing and lifecycle
- Trying to capture the many implications that children's impact on housing demand has
- Aim is policy:
  - Suppose we want to increase fertility
  - How can housing help? What costs?
- More work is needed:
  - Theory: important to understand the role of constraints and sorting incentives. Can we say something about policy design?
  - Empirically: use new geocoded data to track location and moves
  - Quantification: use the model to evaluate policies that target housing to parents, and understand the channels

Thank You

Appendix: Theory Details, PSID Details, and Robustness

Appendix: Entry Accounting Algorithm Back

1. Solve the benchmark in normalized units: $S^* = 1$. The normalized composition gives E^*_{0C}, E^*_{0P}, $E^*_0 = E^*_{0C} + E^*_{0P}$, and B^*_0.
2. Pick the benchmark outside margin. Choose $\bar{W}^E$ so that entry probabilities satisfy
$$q^{E,*} = \pi^{E,*}_C + \pi^{E,*}_P.$$
3. Use the aggregate stationarity equation at $S^* = 1$ to recover the outside-origin potential entrant mass:
$$M = \frac{E^*_0}{q^{E,*}} - B^*_0.$$
4. The solve must also reproduce entry by location:
$$SE_{0i} = [M + SB_0]\,\pi^E_i, \qquad i \in \{C, P\}.$$
The scalar equation for S is the sum of these two conditions.
5. In counterfactuals, keep $(M, \bar{W}^E, \kappa_E)$ fixed. Recompute household values, E_0, B_0, and q^E, then set
$$S = \frac{q^E M}{E_0 - q^E B_0}, \qquad E_i = [M + SB_0]\,\pi^E_i.$$

Appendix: Distributional Stationarity

- The stationary object is the normalized composition g together with city scale S. The level mass of any household state is Sg.

- Incumbent households transition through policies, fertility, location choice, aging, and death.

- Location-specific entrant masses are

$$E_i = [M + SB_0]\, \pi_i^E, \qquad i \in \{C, P\}.$$

  These entrants are placed at the entrant states x_i^E.

- The scalar equation for S is only the sum of the two entry-location equations. The equilibrium still has to reproduce the location distribution inside g.

Calibration Strategy

Externally set

- Lifecycle structure, taxes, interest rate
- Collateral ($\phi = 0.80$), transaction cost ($\psi = 0.06$)
- Housing ladder and supply elasticities

Internally calibrated (SMM)

- 15 parameters: 2 inverted (amenity, rent anchor) + 13 searched
- Fertility, tenure, mobility, saving, bequests
- 17 target moments from ACS and PSID

- Five owner sizes at $\{5.2, 5.85, 6.55, 7.55, 9.0\}$ rooms; rental cap at 4.8 rooms.
- Bequests parameterized by (θ_0, θ_n) with fixed $\theta_1 = 0.01$.

Appendix: Setup — Utility, Demands, Indirect Utility back

Common preliminaries for the derivations that follow.

Utility (CRRA over Cobb–Douglas Stone–Geary):

$$u(c, h; m) = \frac{\left[(c - \bar{c}(m))^{\alpha}(h - \bar{h}(m))^{1-\alpha}\right]^{1-\sigma}}{1-\sigma}.$$

Interior demands (σ-invariant). With M household resources, rent r_i in location i, and $\tilde{M}_i(\bar{h}) = M - \bar{c} - r_i \bar{h}$:

$$c^* - \bar{c} = \alpha\, \tilde{M}_i, \qquad h^* - \bar{h} = (1-\alpha)\, \tilde{M}_i / r_i.$$

Indirect utility with amenity $\mathcal{E}_i$:

$$\mathcal{V}_i(\bar{h}; M) = \frac{Z^*(\bar{h}; M, r_i)^{1-\sigma}}{1-\sigma} + \mathcal{E}_i, \quad Z^* = \alpha^{\alpha}(1-\alpha)^{1-\alpha}\, \tilde{M}_i\, r_i^{-(1-\alpha)}.$$

Appendix: Sorting Derivation back

For $r_C > r_P$, $\mathcal{E}_C > \mathcal{E}_P$, define $\Delta\mathcal{V}(\bar{h}; M) = \mathcal{V}_C(\bar{h}; M) - \mathcal{V}_P(\bar{h}; M)$.

Derivative. With $\gamma \equiv \alpha + \sigma(1-\alpha) > 0$ and $\kappa \equiv \alpha^\alpha(1-\alpha)^{1-\alpha}$,

$$\frac{d\,\Delta\mathcal{V}}{d\bar{h}} = -\,\kappa^{1-\sigma}\left[\frac{r_C^\gamma}{\tilde{M}_C^\sigma} - \frac{r_P^\gamma}{\tilde{M}_P^\sigma}\right] < 0.$$

Sign. $r_C > r_P$ implies $r_C^\gamma > r_P^\gamma$; $\tilde{M}_C < \tilde{M}_P$ implies $\tilde{M}_C^{-\sigma} > \tilde{M}_P^{-\sigma}$. Both factors favour C, so the bracket is strictly positive and the derivative is strictly negative.

Threshold. Since $\Delta\mathcal{V}$ is continuous and strictly decreasing in $\bar{h}$, the equation $\Delta\mathcal{V}(\hat{h}; M) = 0$ has at most one solution. Existence on a non-degenerate interval follows from $\Delta\mathcal{V}(0; M) > 0$ (amenity-positive) and $\Delta\mathcal{V}(\bar{h}; M) \to -\infty$ as $\bar{h} \uparrow (M - \bar{c})/r_C$ (infeasibility in C). Unique threshold $\hat{h}(M)$.

Parent–childless ordering. $\bar{h}(1) > \bar{h}(0)$ and $\Delta\mathcal{V}$ strictly decreasing $\Rightarrow$ parents weakly prefer the periphery, strictly when $\bar{h}(0) < \hat{h}(M) < \bar{h}(1)$.

Appendix: Elasticity-Attenuation Derivation

back

Simplified demand (interior renter at rent r; assume $\bar{c}(0) = \bar{c}(1) = \bar{c}$ so that only the housing requirement varies with the number of children):

$$h^*(n) = \alpha\,\bar{h}(n) + (1-\alpha)\,\frac{M-\bar{c}}{r}.$$

Elasticities.

$$e_M(n) = \frac{\partial \log h^*}{\partial \log M} = \frac{(1-\alpha)(M-\bar{c})}{r\,h^*(n)}, \qquad e_r(n) = \frac{\partial \log h^*}{\partial \log r} = -\frac{(1-\alpha)(M-\bar{c})}{r\,h^*(n)}.$$

Attenuation. $h^*(n)$ is strictly increasing in $\bar{h}(n)$ (coefficient $\alpha > 0$); the numerator is independent of $\bar{h}(n)$. Hence both $e_M(n)$ and $|e_r(n)|$ are strictly decreasing in $\bar{h}(n)$. Since $\bar{h}(1) > \bar{h}(0)$:

$$e_M(1) < e_M(0), \qquad |e_r(1)| < |e_r(0)|.$$

Limits. As $\bar{h} \to 0$, both elasticities $\to \mp 1$ (Cobb–Douglas benchmark). As $\bar{h} \to (M-\bar{c})/r$ (feasibility boundary), both $\to 0$ (pure subsistence).

Identification. The gap $e_M(0) - e_M(1) > 0$ pins down $\Delta\bar{h} = \bar{h}(1) - \bar{h}(0)$ given α, $M-\bar{c}$, and r.

Appendix: Cross-Metro Sorting-Wedge Derivation

back

Indirect utility on an interior renter branch in location i with n children ($\sigma = 1$ for transparency; signs go through for any $\sigma > 0$):

$$\mathcal{V}_i(n; M) = \log \tilde{M}_i^n - (1-\alpha)\log r_i + \mathcal{E}_i, \quad \tilde{M}_i^n = M - \bar{c}(n) - r_i \bar{h}(n).$$

Sorting wedge. Define $\Delta\mathcal{V}(n; M) = \mathcal{V}_C(n; M) - \mathcal{V}_P(n; M)$ and $\mathcal{W}(M) \equiv \Delta\mathcal{V}(0; M) - \Delta\mathcal{V}(1; M)$:

$$\mathcal{W}(M) = \log \frac{\tilde{M}_C^0\, \tilde{M}_P^1}{\tilde{M}_C^1\, \tilde{M}_P^0}.$$

Comparative statics in rents. Using $\partial \tilde{M}_i^n / \partial r_i = -\bar{h}(n)$,

$$\frac{\partial \mathcal{W}}{\partial r_C} = \frac{\delta_h (M - \bar{c}(0)) + \bar{h}(0)\,\delta_c}{\tilde{M}_C^1\, \tilde{M}_C^0} > 0, \qquad \frac{\partial \mathcal{W}}{\partial r_P} = -\frac{\delta_h (M - \bar{c}(0)) + \bar{h}(0)\,\delta_c}{\tilde{M}_P^1\, \tilde{M}_P^0} < 0.$$

Numerators positive by interior feasibility ($M > \bar{c}(0)$) plus $\delta_c, \delta_h, \bar{h}(0) > 0$; denominators positive on the interior renter branch.

Reading. Raising the centre rent (or lowering the periphery rent) widens the parent–childless sorting wedge: the cross-metro elasticity of sorting strength to the local rent gap is strictly positive. Same algebraic structure carries through for capped renters (replace $\tilde{M}_i^n$ with $\check{M}_i^n = M - r_i \bar{h}_R - \bar{c}(n)$).

Appendix: Constrained-Squeeze CEV Derivation back

CEV solves $u(c, h, 0) = u(c + \mathrm{CEV}, h, 1)$: housing quantity is held fixed at h, and the post-child surplus shrinks via $\bar{h}(n)$ from $s^0 = h - \bar{h}(0)$ to $s^0 - \Delta\bar{h} = h - \bar{h}(1)$. Under Stone–Geary $u = [(c - \bar{c}(n))^\alpha (h - \bar{h}(n))^{1-\alpha}]^{1-\sigma}/(1-\sigma)$ and $\bar{c}(0) = \bar{c}(1) = \bar{c}$, equating utilities:

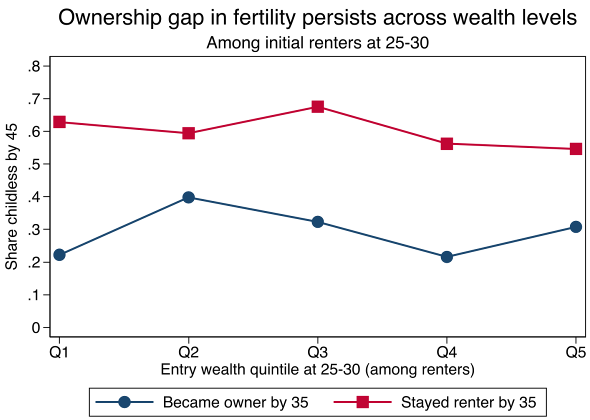
$$(c - \bar{c})^\alpha \, (s^0)^{1-\alpha} = (c + \mathrm{CEV} - \bar{c})^\alpha \, (s^0 - \Delta\bar{h})^{1-\alpha}.$$

Rearrange:

$$\frac{c + \mathrm{CEV} - \bar{c}}{c - \bar{c}} = \left(\frac{s^0}{s^0 - \Delta\bar{h}}\right)^{(1-\alpha)/\alpha} \implies \mathrm{CEV} = (c - \bar{c})\left[\left(\tfrac{s^0}{s^0 - \Delta\bar{h}}\right)^{(1-\alpha)/\alpha} - 1\right].$$

Ownership Is the Binding Constraint on Fertility

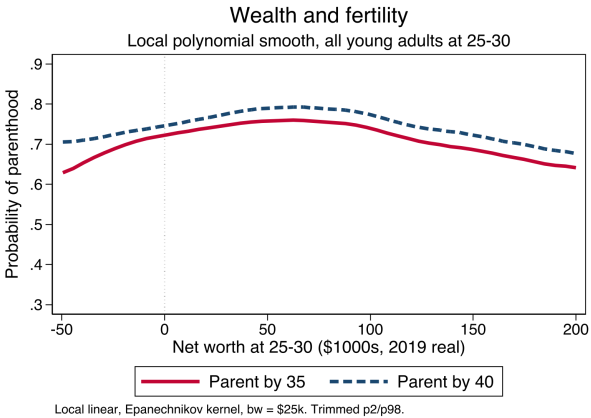
Aggregate Gap



Sample: PSID renters at ages 25–30, pre-first-birth, followed to age 45 ($N = 1{,}127$).

- At **every** wealth quintile, renters who transition to ownership by 35 are 20–35 pp less likely to remain childless by 45.
- The gap does not shrink with wealth: even wealthy renters who do not buy remain $\sim$55% childless.
- Mediation regression: wealth has **zero** direct effect on fertility ($p = 0.92$): the entire effect runs

Wealth and Fertility: Two Channels

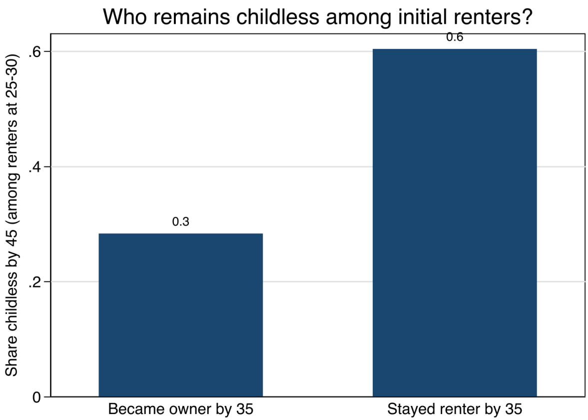


The unconditional wealth–fertility relationship is **hump-shaped**:

- **Rising part** (NW $<$ \$80k): more wealth $\Rightarrow$ can clear down payment $\Rightarrow$ access family-sized housing $\Rightarrow$ have children. The *resource/housing channel*.
- **Falling part** (NW $>$ \$80k): wealthier young adults are disproportionately high human capital and career-invested. The *opportunity cost channel*.

The non-monotonicity is why we need

Aggregate Ownership Transition Gap Back



Among renters at 25–30 who have not yet had a first child:

- Those who transition to ownership by 35: **28%** childless at 45.
- Those who stay renters through 35: **60%** childless at 45.
- A **32 pp** extensive-margin gap—larger than any within-tenure wealth effect.

Tenure status averaged over ages 25–30 (renter) and 31–35 (transition outcome). Childlessness measured by 45 using reported age at first birth.

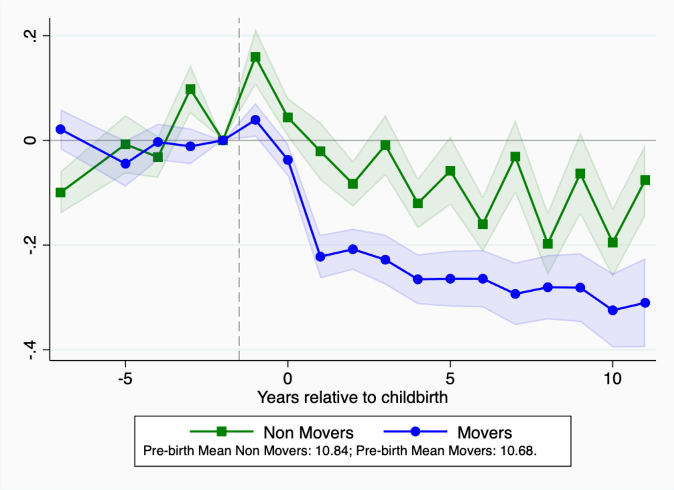
PSID: Housing Also Expands After Second Birth

Back

Baseline second-birth event study with stay-at-one-child controls: the +3 coefficient is about **0.57 rooms**.

PSID: Income Trajectories, Movers vs Non-Movers

No IDFE



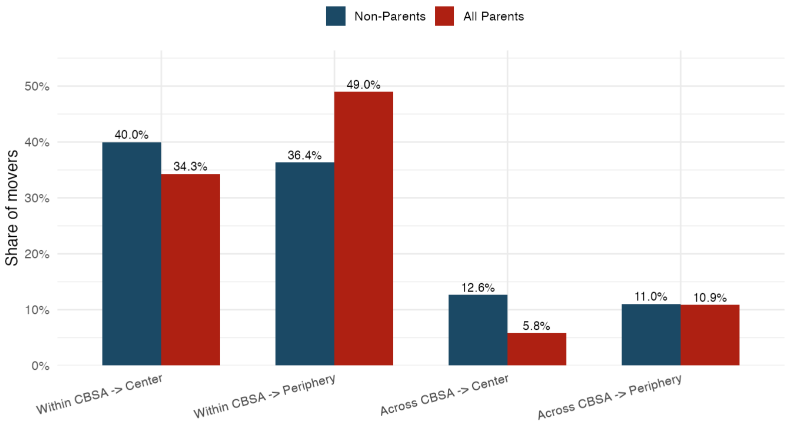
Defining Center and Periphery Within Large Metros

Back

Sample: U.S. metros with 2000 population > 1M (51 metros). Core = tracts containing the closest 30% of metro population to the CBD. A PUMA is classified as center if at least 10% of its population lives in core tracts. Center pop share: 45.0%.

Most Adjustment Is Within Metros

Back



- Most adjustment happens **within** metros rather than across metros.

Mover Regressions: Parent vs. Non-Parent

Back

Outcome (pp)	(1) New Parent	(2) Any Parent
Across-CBSA move	-4.56^{***}	-5.17^{***}
	(0.78)	(0.54)
Center destination $\mid$ within-CBSA	-11.21^{***}	-11.44^{***}
	(0.66)	(0.53)
Within CBSA $\rightarrow$ Center	-6.77^{***}	-6.79^{***}
	(0.77)	(0.65)
Within CBSA $\rightarrow$ Periphery	11.33^{***}	11.95^{***}
	(0.49)	(0.30)
Center dest. $\mid$ center origin	-9.69^{***}	-8.91^{***}
	(0.81)	(0.53)
Center dest. $\mid$ periphery origin	-10.71^{***}	-11.32^{***}
	(0.51)	(0.47)
Age and year FE	Yes	Yes
Sex and race controls	Yes	Yes

Notes. OLS on ACS 2012–2023 within-metro movers, ages 22–45, weighted by `perwt`. Coefficients in percentage points. Column (1) uses a *new-parent* indicator (eldest child < 4); column (2) uses an *any-parent* indicator. Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Cross-Metro Sorting Regression

Back

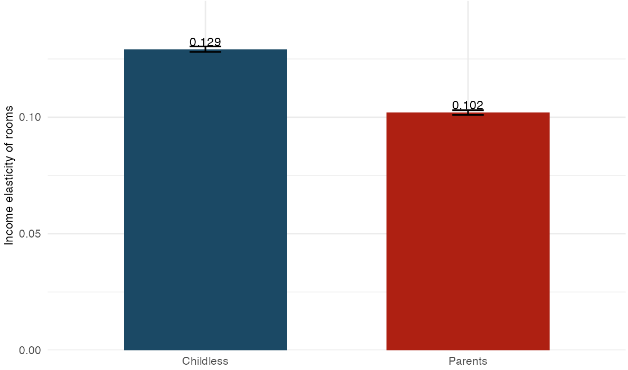
Dependent gap (pp)	Slope on log rent gap	Metros
Center-share gap	6.23	42
	(5.19)	
Center-destination gap | within-CBSA movers	23.53***	42
	(6.18)	

Notes. Unit of observation is the metro. Each dependent variable is the non-parent minus new-parent gap in percentage points. Regressor is the local log center–periphery rent-per-room gap. Each regression is weighted by its corresponding metro sample size. *** $p < 0.01$.

Housing Demand Elasticity Among Renters

Back Regression

Model: parents less income-elastic in housing ($\bar{h}(1) > \bar{h}(0)$).



Mean rooms among renters: 4.20 for childless households, 4.92 for parents.

- Parent renters start from a larger space baseline, but rooms rise less with income: elasticity 0.102 for parents versus 0.129 for childless renters.

Renter Housing-Demand Elasticity

Back

Outcome	Childless slope	Parent interaction	Parent slope
Log rooms	0.129***	−0.027***	0.102***
	(0.001)	(0.001)	(0.001)
Metro FE	Yes	Yes	Yes
Age and year FE	Yes	Yes	Yes
Sex and race controls	Yes	Yes	Yes
Observations		3,024,081	

Notes. OLS on ACS renters, ages 22–45, 2012–2023, weighted by `perwt`. Specification is log(rooms) on log(income), parent status, and the interaction, with metro, age, and year fixed effects plus sex and race controls. Standard errors in parentheses. *** $p < 0.01$.

Potential Reasons to Move in the PSID Questionnaire

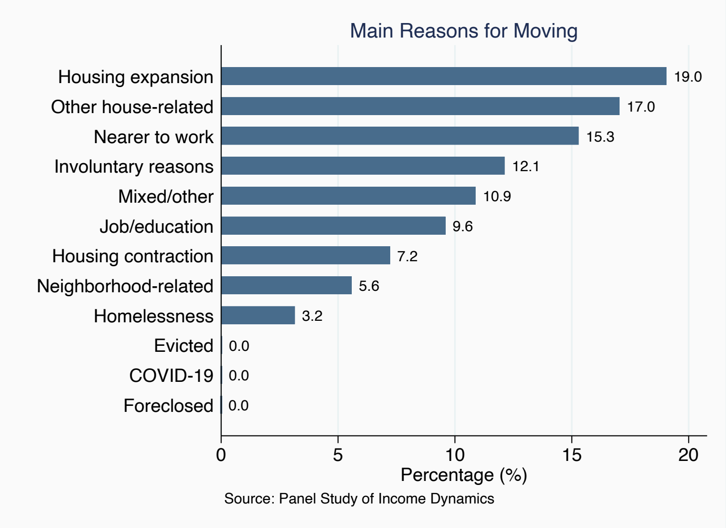
Back

Movers are asked: "What is the main reason you moved?". Categories:

- **Purposive productive:** take another job; transfer; stopped going to school.
- **Nearer to work.**
- **Purposive consumptive — expansion of housing:** more space; higher rent; better place.
- **Purposive consumptive — contraction of housing:** less space; less rent.
- **Purposive consumptive — other house-related:** get own home; got married; physical condition of prior unit.
- **Purposive consumptive — neighborhood:** better neighborhood; schooling; closer to friends/relatives.
- **Involuntary:** housing unit coming down; armed services; health; divorce; forced retirement.
- **Ambiguous or mixed:** save money, neighbors moved away, retiring.
- Became homeless or no longer homeless; COVID-19; evicted / foreclosed.

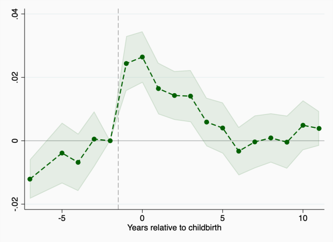
Distribution of Reasons to Move Among Non-Parents

This tabulates values only for non-parents (both “never” and “not yet”).

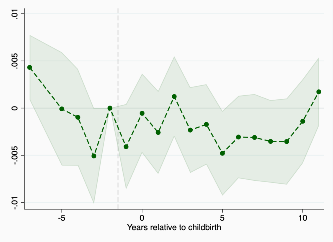


Event Studies: Housing and Location Reasons

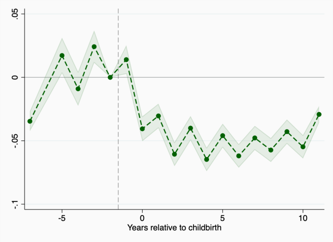
Back



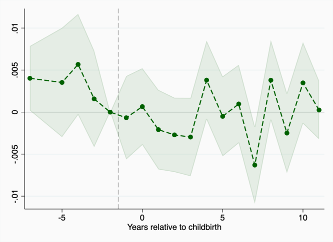
Expansion of housing



Contraction of housing

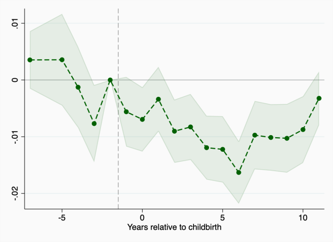


Other house-related

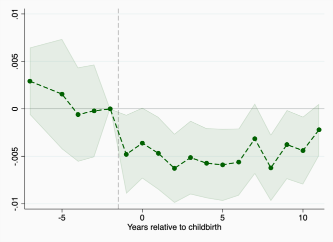


Neighborhood

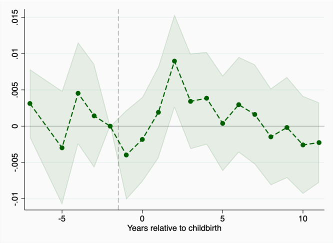
Event Studies: Work and Involuntary Reasons



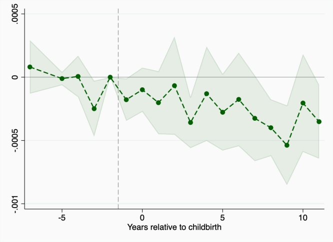
Productive reasons



To get nearer to work

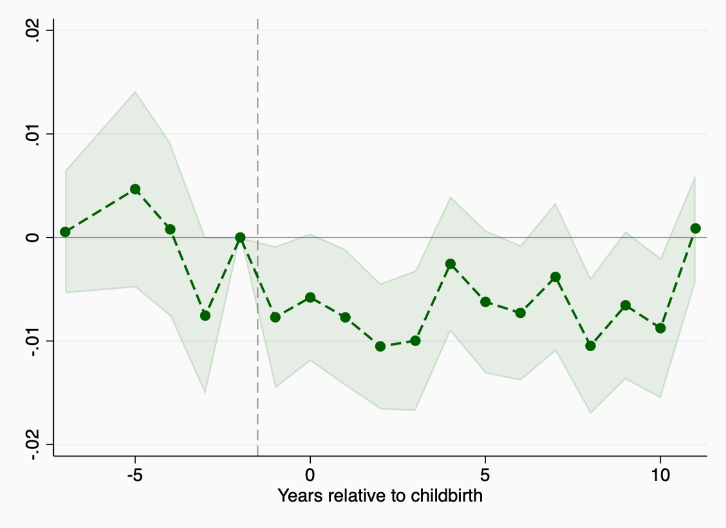


Involuntary reasons



Was evicted

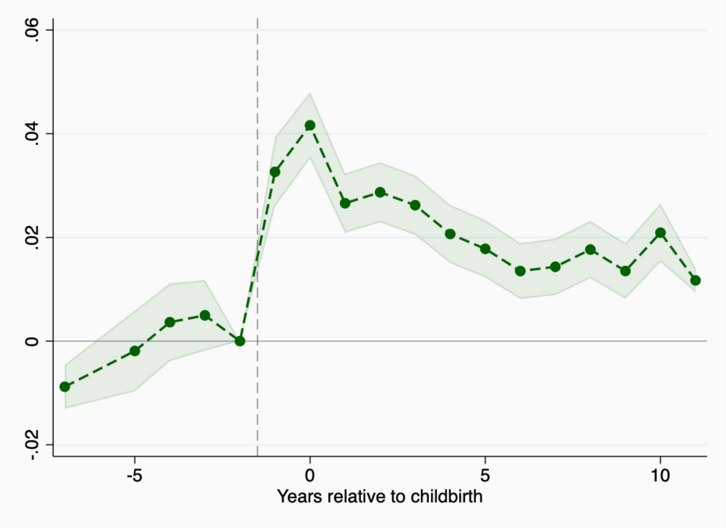
Event Studies: Other Reasons



Mixed or other reasons

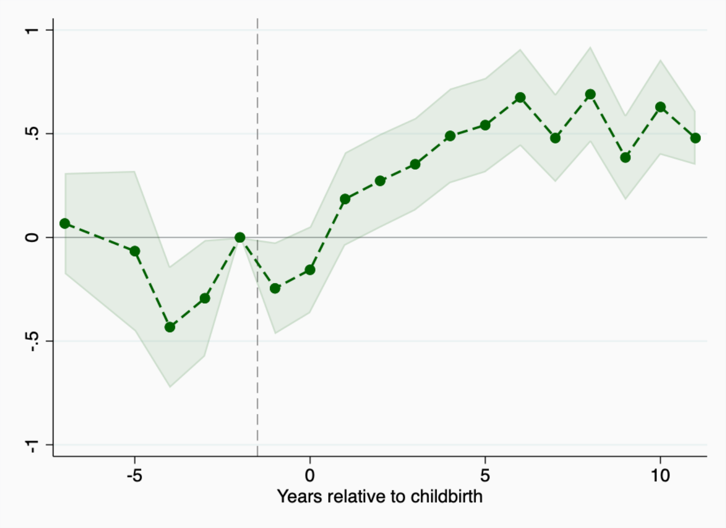
No Individual FE: Reasons to Move (Expansion of Housing)

Back



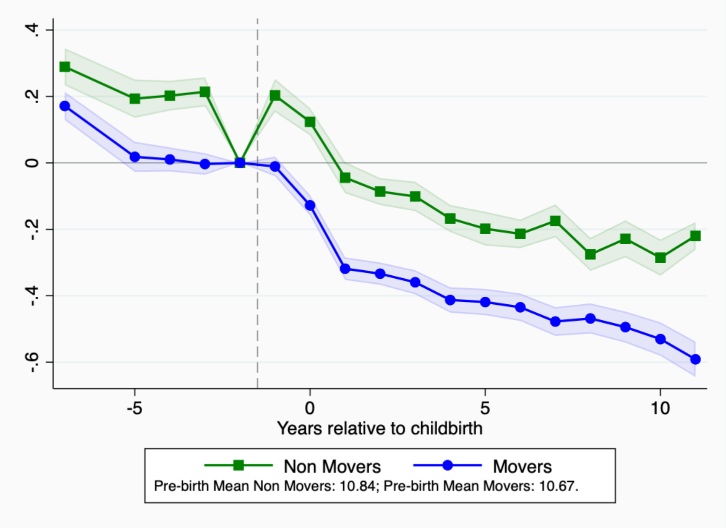
No Individual FE: Number of Rooms Around First Birth

Back



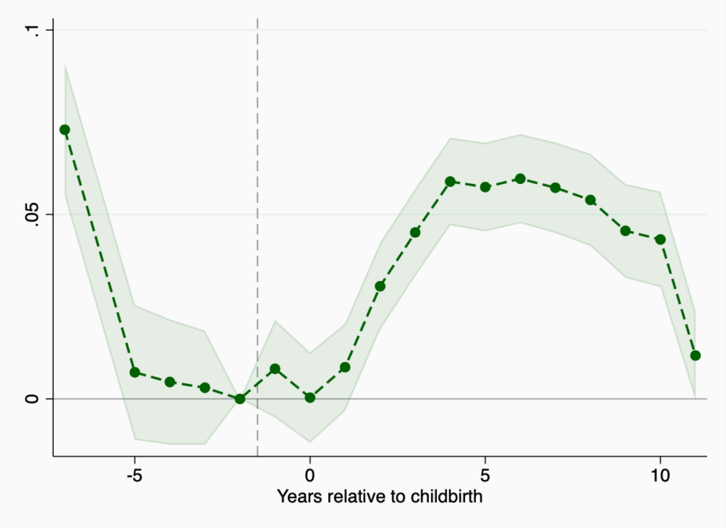
No Individual FE: Family Income Around First Birth

Back



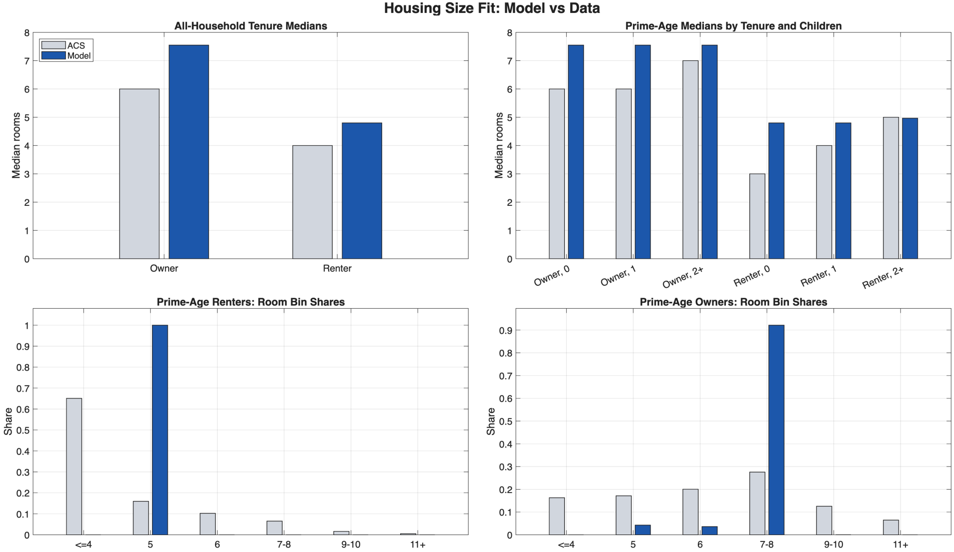
No Individual FE: Homeownership Around First Birth

Back



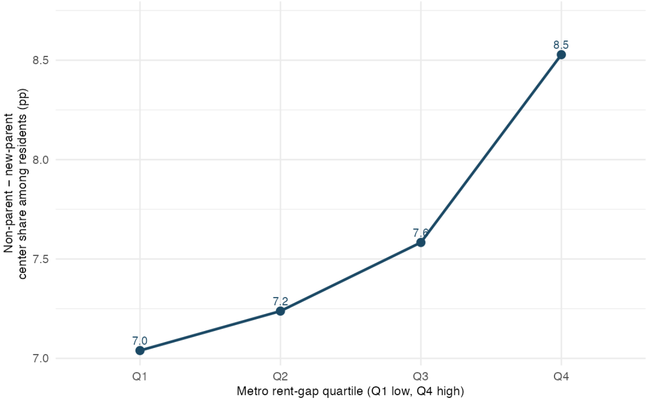
Housing Size Fit: Model vs Data

Back



Sorting Strength: Residents (Stock) Back

Model: parent share falls with local rent; $\partial \mathcal{W} / \partial r_C > 0$.



Y-axis: non-parent minus new-parent center share among all residents (pp). Base: 49.4% for non-parents, 41.6% for new parents.

- Stock-based sorting gap also rises with the rent gap, but the gradient is flatter than among movers.